

Anbang Wang

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 Wang Anbang |  Aawangas

Hong Kong SAR, China

RESEARCH INTEREST

My research interests primarily focus on computer vision and machine learning. Specifically, I am particularly interested in developing fundamental methods to fully leverage large-scale world knowledge. My recent interests are particularly centered on 3D vision.

EDUCATION

- **The Hong Kong University of Science and Technology** 2023/9 - 2027/9 (expected)
BSc in data science and technology
◦ GPA: 4.035/4.3 (top 2%)
Hong Kong SAR, China
- **École Polytechnique Fédérale de Lausanne** 2026/2 - 2026/7 (expected)
Exchange Program in Computer Science section
Lausanne, Switzerland

PUBLICATIONS

- **SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation** *arXiv*
Anbang Wang*, Yuzhuo Ao*, Shangzhe Wu, Chi-Keung Tang
* Equal contribution.
- **ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation** *arXiv*
Yuzhuo Ao*, Anbang Wang*, Yu-Wing Tai, Chi-Keung Tang
* Equal contribution.
- **MedSapiens: Taking a Pose to Rethink Medical Imaging Landmark Detection** *MedIA 2026*
Marawan Elbatel*, Anbang Wang*, Keyuan Liu, Kaouther Mouheb, Enrique Almar-Munoz, Lizhuo Lin, Yanqi Yang, Karim Lekadir, Xiaomeng Li
* Equal contribution.
- **Geometric-Guided Few-Shot Dental Landmark Detection with Human-Centric Foundation Model** *MICCAI 2025*
Anbang Wang*, Marawan Elbatel*, Keyuan Liu, Lizhuo Lin, Meng Lan, Yanqi Yang, Xiaomeng Li
* Equal contribution.

PROJECTS

- **SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation** 2026/11 - 2026/03
Advised by Prof. Chi-Keung Tang and Prof. Shangzhe Wu 
 - Developed SK-Adapter, a lightweight skeleton-based adapter that injects joint-coordinate and topology tokens into a frozen native 3D generation backbone via cross-attention.
 - Introduced skeletons as first-class structural control signals for precise 3D asset generation, reducing ambiguity from text- or image-only prompts.
 - Contributed to Objaverse-TMS, a 24k text-mesh-skeleton dataset, and demonstrated robust skeleton-conditioned generation and local 3D editing while preserving geometry and texture quality.
- **ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation** 2025/06-2025/09
Advised by Prof. Chi-Keung Tang And Prof. Yu-Wing Tai 
 - Developed ReasonNavi, a framework that leverages MLLMs for global reasoning and a deterministic planner for local navigation, creating a novel reason-then-act paradigm.
 - Enabled unified, zero-shot navigation across diverse tasks (object, image, text goals), eliminating the need for task-specific fine-tuning or reinforcement learning.
 - Demonstrated superior performance over existing methods in efficiency and reliability.
- **Geometric-Guided Few-Shot Dental Landmark Detection with Human-Centric Foundation Model** 2024/12 - 2025/03
Advised by Prof. Xiaomeng Li 
 - Engineered a novel few-shot learning framework (GeoSapiens) by creatively adapting a human-centric foundation model (Sapiens) for a challenging medical imaging task.
 - Innovated a novel geometric loss function that encodes anatomical structural priors, significantly improving the model's robustness and precision in low-data scenarios.
 - Achieved state-of-the-art (SOTA) results, outperforming the previous leading method by 8.18% in Success Detection Rate (SDR) at the clinically-critical 0.5mm precision threshold.
- **MedSapiens: Taking a Pose to Rethink Medical Imaging Landmark Detection** 2024/11 - 2025/11
Advised by Prof. Xiaomeng Li 

- Developed MedSapiens by adapting a large-scale human-centric foundation model (Sapiens) for medical anatomical landmark detection, leveraging its inherent spatial pose localization priors across diverse imaging modalities.
- Harmonized heterogeneous datasets across four distinct anatomical regions (head, hand, chest, and legs) into a unified training pipeline, utilizing LoRA and heatmap-based decoding to capture shared spatial hierarchies across diverse clinical tasks.
- Established new state-of-the-art (SOTA) performance, outperforming generalist models by 5.26% and specialist models by 21.81% in Success Detection Rate (SDR), while achieving a 2.69% improvement in few-shot generalization on novel tasks.
- **Unsupervised Domain Adaptation for 3D Bone Reconstruction from 2D X-rays** 2024/09 - 2024/11
Advised by Prof. Xiaomeng Li
 - **Investigated** the challenge of domain shift in 3D bone reconstruction by building a baseline model utilizing the Pixel-Aligned Implicit Function (PiFU) framework.
 - **Implemented** a state-of-the-art unsupervised domain adaptation (UDA) algorithm to enhance the model's generalization capabilities across different X-ray datasets.
 - **Aimed to improve** reconstruction accuracy and robustness on out-of-domain data without requiring costly new annotations from the target domain.
- **Knowledge Distillation on Skin Lesion Segmentation** 2024/01 - 2024/09
Advised by Prof. Albert C. S. Chung 
 - **Developed** a lightweight segmentation model for skin lesions by designing and implementing a knowledge distillation (KD) pipeline.
 - **Engineered** a student-teacher framework where knowledge from a large, complex model was effectively transferred to a compact student network.

HONORS AND AWARDS

- HKUST Scholarship for Continuing Undergraduate Students (highest band)
- UROP research travel sponsorship
- Dean's List for all semesters

TEACHING EXPERIENCE

- **Teaching Assistant of COMP 2711, Discrete Math and its Applications in Computer Science** 2024/09 - 2024/12
HKUST, CSE department
 - **Assist** weekly tutorial sessions for undergraduate students, clarifying core concepts in logic, set theory, graph theory, and combinatorics.
 - **Provided** regular one-on-one guidance, helping students debug complex proofs and master challenging assignment problems.

ACADEMIC ENGAGEMENT AND SERVICE

- **Reviewer**, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- **Reviewer**, European Conference on Computer Vision (ECCV), 2026
- **Attendee**, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), , Daejeon, Republic of Korea, Oct 2025

SKILLS

- **Programming Languages:** Python (Proficient), C/C++ (Proficient)
- **Frameworks & Libraries:** PyTorch, NumPy, Pandas, Scikit-learn, OpenCV
- **Developer Tools & OS:** Git, GitHub, Linux, LaTeX
- **Languages:** Mandarin (Native), English (Fluent - IELTS 7.5/9.0)